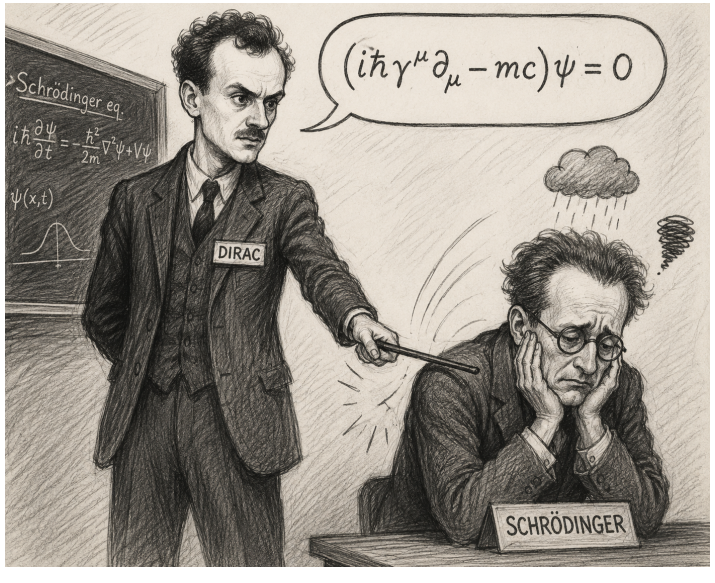
Relativistic Effects: A "Gentle" Introduction

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Why this seminar?

About me

- PhD in Toulouse under the supervision of Trond Saue.
- Research project developed along two main directions:
  - ⋆ Development of relativistic CCSDT and CCSDTQ methods.
  - ⋆ Accurate extraction of nuclear quadrupole moments (NQMs).

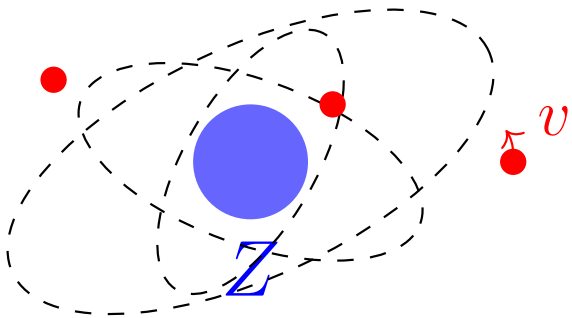
My PhD thesis (fun included)

Relativistic effects

- Electrons in the vicinity of heavy nuclei move at relativistic speeds.
  - ⋆ Scalar relativistic effects
  - ⋆ Spin–orbit coupling
- Non-relativistic quantum mechanics is no longer sufficient.

Relativistic effects

Scalar effects



The Lorentz factor

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \qquad \begin{cases} v & \text{particle speed} \\ c & \text{speed of light} \end{cases} \tag{1}$$

Scalar relativistic effects

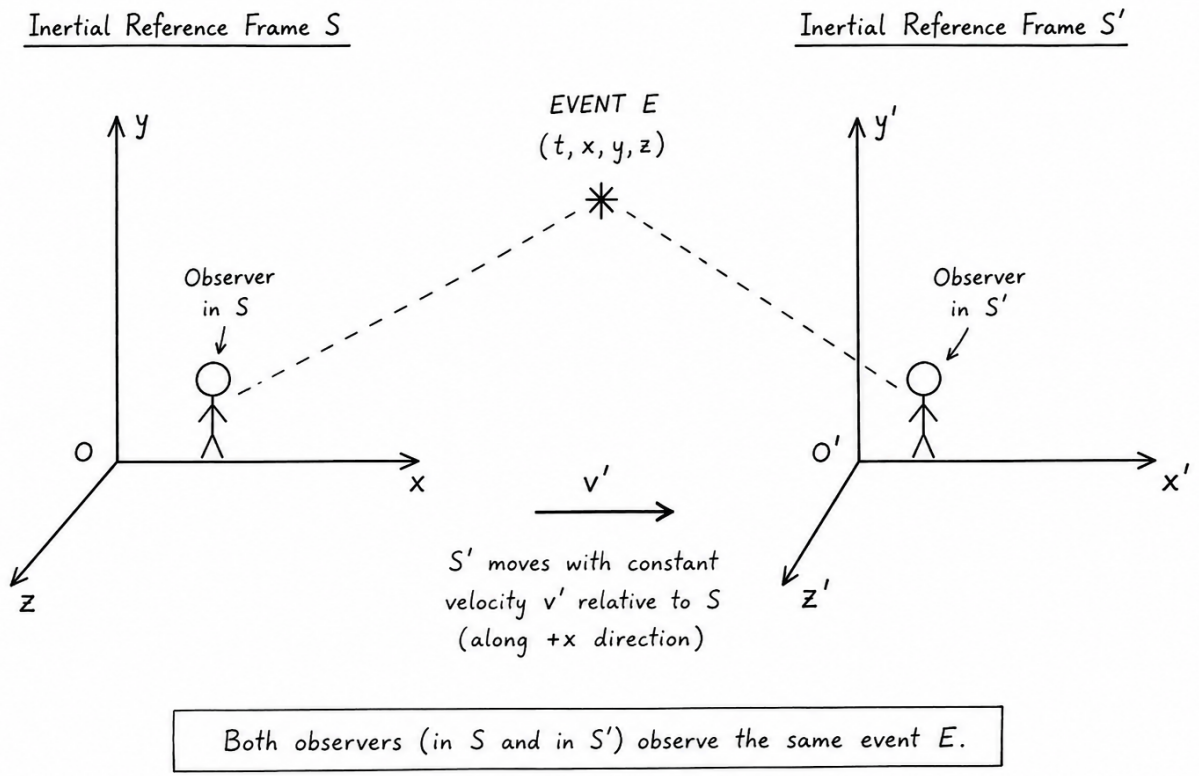
- In a.u. $v_{1s} = Z$
- The relativistic mass is $m_{\text{rel}} = \gamma m_{\text{e}}$
- The Bohr radius is $a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m}$

Direct/Indirect effects

- $\langle r \rangle_s^{\text{rel}}, \langle r \rangle_p^{\text{rel}} < \langle r \rangle_s^{\text{non-rel}}, \langle r \rangle_p^{\text{non-rel}}$
- $\langle r \rangle_d^{\text{rel}}, \langle r \rangle_f^{\text{rel}} > \langle r \rangle_d^{\text{non-rel}}, \langle r \rangle_f^{\text{non-rel}}$

Interruption: Lorentz transformation

Lorentz transformations relate the space-time coordinates of two inertial reference frames, ensuring that the laws of physics and the speed of light remain invariant in both frames.



Lorentz (1904)

$$\mathbf{r}' = \mathbf{r} + \frac{\gamma - 1}{v^2}(\mathbf{v} \cdot \mathbf{r})\mathbf{v} - \gamma \mathbf{v} t, \quad t' = \gamma\left(t - \frac{\mathbf{v} \cdot \mathbf{r}}{c^2}\right) \tag{2}$$

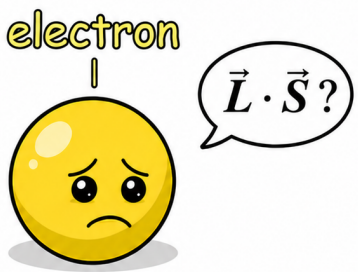
Relativistic effects

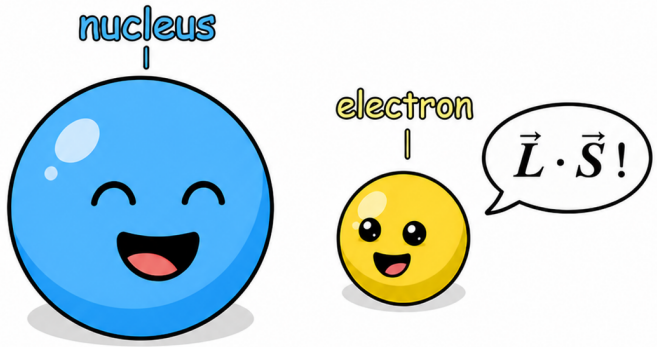
Spin-orbit interaction effect (1)

First clarification

The spin–orbit interaction IS NOT a direct interaction between the electron spin and its orbital angular momentum!

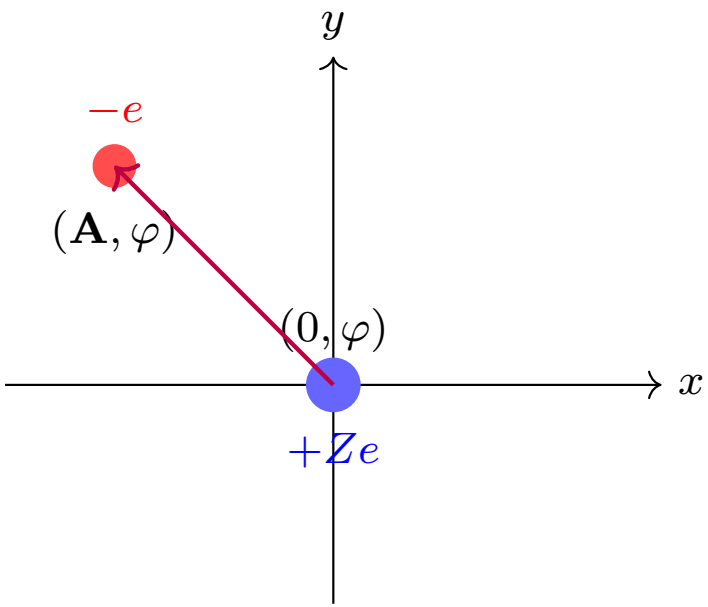
An electron **alone** does not experience any spin-orbit interaction!
Spin–orbit interaction arises **only** in the presence of an **electric field**.





Relativistic effects

Spin-orbit interaction effect (2)



- In the nucleus frame:
  - $\mathbf{B} = 0$
  - only $\mathbf{E} = \frac{Ze}{4\pi\varepsilon_0}\frac{\mathbf{r}}{r^3}$
- In the instantaneous electron frame (Lorentz transformation):

$$\mathbf{E}' = \gamma\left(\mathbf{E} + \mathbf{v}\times\mathbf{B}\right), \quad \mathbf{B}' = \gamma\left(\mathbf{B} - \frac{1}{c^2}\mathbf{v}\times\mathbf{E}\right)$$

Spin-orbit coupling is the interaction of the electron spin with the magnetic field induced by a charge in relative motion

$$H_{\mathrm{SO}} \propto \mathbf{S}\cdot\mathbf{B}' = \mathbf{S}\cdot\mathbf{L}$$

⋆ $[H_{\mathrm{SO}}, L^2] \neq 0, \quad [H_{\mathrm{SO}}, S^2] \neq 0$

⋆ $\mathbf{J} = \mathbf{L} + \mathbf{S} \rightarrow [H_{\mathrm{SO}}, J^2] = 0$

⋆ It gives rise to the fine structure ($j = \ell \pm 1/2$).

Building the relativistic Hamiltonian

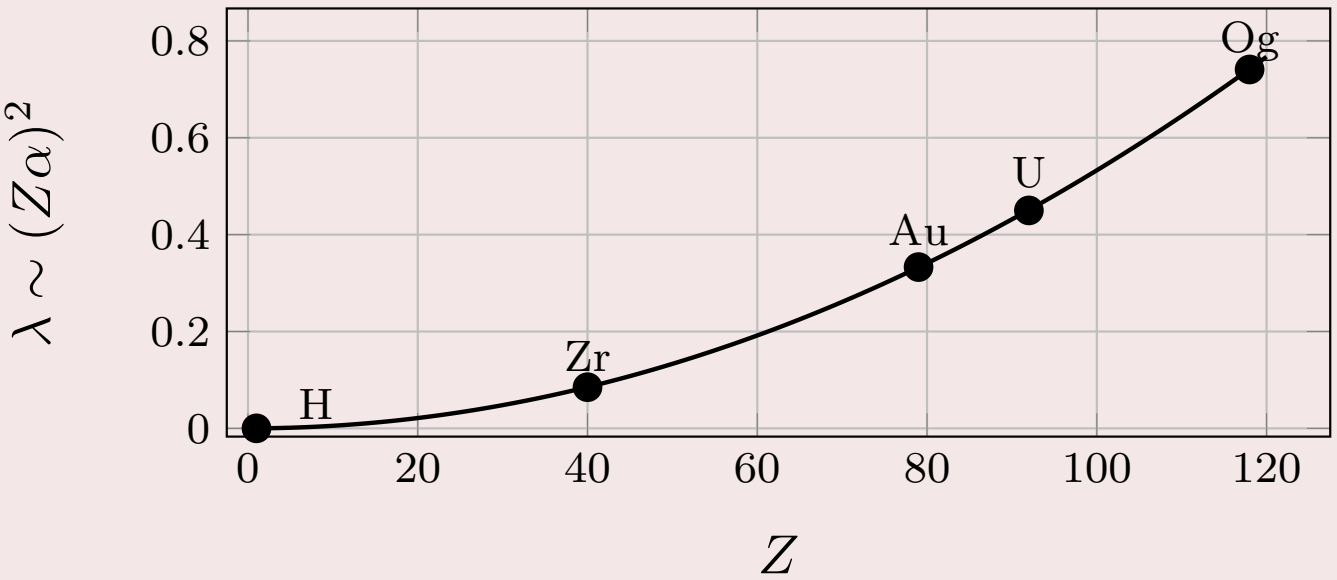
Relativistic effects are often introduced as corrections to the nonrelativistic Hamiltonian

$$H = H^{\mathrm{NR}} + \lambda H^{\mathrm{rel}} \tag{3}$$

Limits of perturbation theory

$$H = H_0 + \lambda V, \quad \lambda \ll 1$$

$$\lambda \sim (Z\alpha)^2 \;\Rightarrow\; \lambda \not\ll 1 \text{ for heavy atoms}$$



Relativistic one-electron part

Classical description

Einstein (1906)

$$E^2 = m^2c^4 + c^2p^2 \tag{4}$$

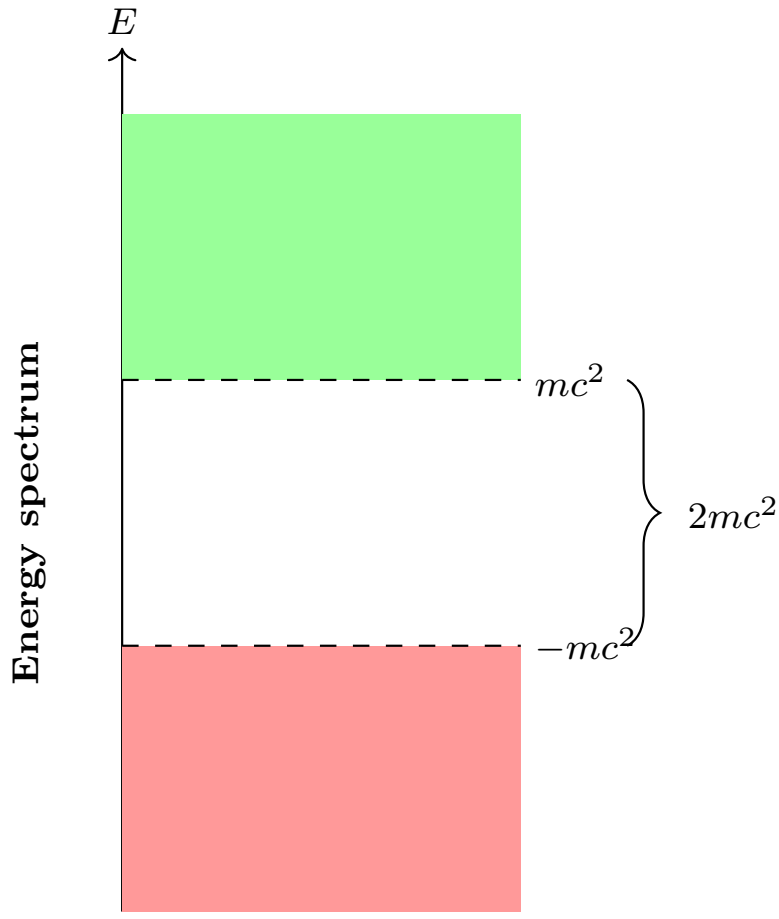
The formal solution leads

$$E = \pm\sqrt{m^2c^4 + c^2p^2}$$

$$E \in (-\infty, -mc^2] \cup [mc^2, +\infty)$$

In classical mechanics, the energy changes continuously.

$$E(t + \mathrm{d}t) = E(t) + \mathrm{d}E$$



Relativistic one-electron part

Why does our beloved Schrödinger equation fail?

Schrödinger (1925)

$$i\hbar\frac{\partial}{\partial t}\psi = \left(-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right)\psi \tag{5}$$

- ⋆ First derivative in time and second derivative in space...

Under a Lorentz transformation:

$$\frac{\partial}{\partial t} = \gamma\frac{\partial}{\partial t'} - \gamma(\mathbf{v}\cdot\nabla'), \tag{6}$$

$$\nabla^2 = \nabla'\cdot\nabla' + \frac{\gamma-1}{v^2}(\mathbf{v}\cdot\nabla')^2 - \frac{2\gamma}{c^2}\,\mathbf{v}\cdot\nabla'\,\frac{\partial}{\partial t'} + \frac{\gamma^2 v^2}{c^4}\,\frac{\partial^2}{\partial t'^2} \tag{7}$$

Failures of the Schrödinger equation

- Space and time derivatives mix under boosts
- Schrödinger equation is NOT Lorentz invariant.

Relativistic one-electron part

First attempt for a quantum mechanical description

Klein–Gordon (1926)

Simple idea: quantize a classical relativistic equation

$$E^2 = p^2c^2 + m^2c^4 \tag{8}$$

Apply canonical quantization:

$$E \to i\hbar\frac{\partial}{\partial t}, \qquad \mathbf{p} \to -i\hbar\nabla$$

This leads to the Klein–Gordon equation:

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2 + \frac{m^2c^2}{\hbar^2}\right)\psi = 0 \tag{9}$$

Failures of the Klein–Gordon equation

- We must deal with the presence of negative-energy solutions
- The probability density is not positive definite

$$\rho_+ = \frac{|E|}{mc^2}|\psi|^2, \qquad \rho_- = -\frac{|E|}{mc^2}|\psi|^2 \tag{10}$$

Relativistic one-electron part

The Dirac equation

Dirac (1928)

$$\left(\hat{h}_0 - i\frac{\partial}{\partial t}\right)\psi = 0 \tag{11}$$

The relativistic free-particle hamiltonian is

$$\hat{h}_0 = c(\mathbf{p}\cdot\boldsymbol{\alpha}) + \beta mc^2 = \begin{pmatrix} mc^2 & c(\mathbf{p}\cdot\boldsymbol{\sigma}) \\ c(\mathbf{p}\cdot\boldsymbol{\sigma}) & -mc^2 \end{pmatrix}$$

- ⋆ Linear in time$\rightarrow \rho \geq 0$.
- ⋆ Spin emerges naturally $\rightarrow \mathbf{S}_i = \frac{\hbar}{2}\boldsymbol{\sigma}_i$.
- ⋆ $\hat{h}_0$ acts on 4-component spinors.

The time-independent equation reads

$$\begin{bmatrix} h_{\mathrm{LL}} & h_{\mathrm{LS}} \\ h_{\mathrm{SL}} & h_{\mathrm{SS}} \end{bmatrix} \begin{bmatrix} \psi_{\mathrm{L}} \\ \psi_{\mathrm{S}} \end{bmatrix} = \varepsilon \begin{bmatrix} \psi_{\mathrm{L}} \\ \psi_{\mathrm{S}} \end{bmatrix} \tag{12}$$

The solutions are 4-component spinors

$$\psi = \begin{bmatrix} \psi_{\mathrm{L}} \\ \psi_{\mathrm{S}} \end{bmatrix} = \begin{bmatrix} \psi_{\mathrm{L}}^{\alpha} \\ \psi_{\mathrm{L}}^{\beta} \\ \psi_{\mathrm{S}}^{\alpha} \\ \psi_{\mathrm{S}}^{\beta} \end{bmatrix} \tag{13}$$

ψ_{L} and ψ_{S} are connected

$$\psi_{\mathrm{S}} = \frac{c(\boldsymbol{\sigma}\cdot\mathbf{p})}{\varepsilon + mc^2}\psi_{\mathrm{L}} \tag{14}$$

- ⋆ Kinetic-balance prescription.

Relativistic one-electron part

Negative-energy solutions

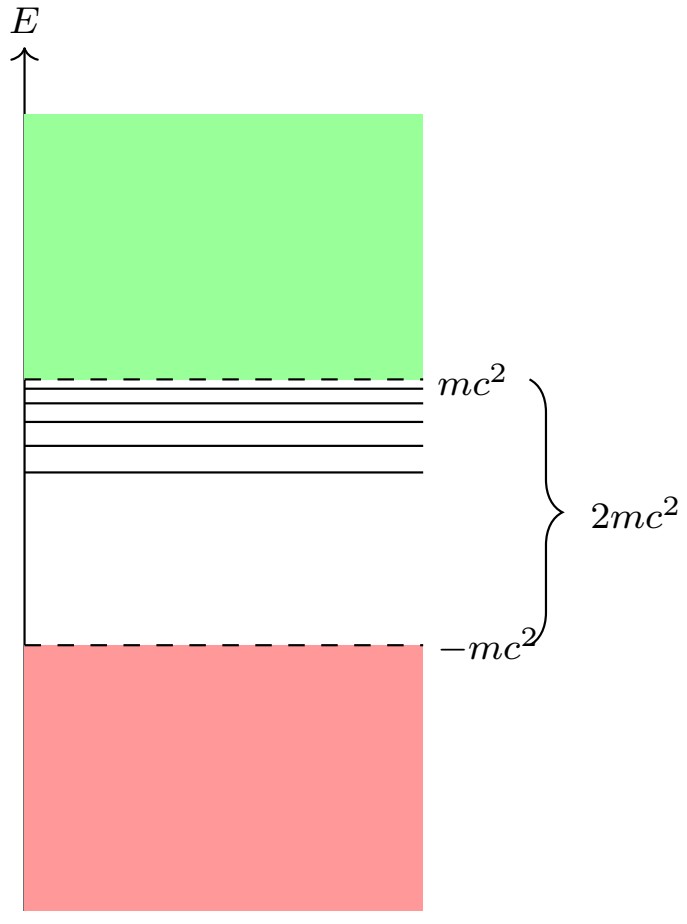
P. A. M. Dirac, *A Theory of Electrons and Protons*, Proc. Roy. Soc. A **126**, 360 (1930).

The Dirac sea

- All NE solutions are occupied.
- The Pauli exclusion principle then hinders electrons from descending into the NE branch.
- The excitation of an electron from the NE band leaves a hole of positive charge.
- Dirac interpreted the hole as the antiparticle of the electron.

C. D. Anderson, *The Positive Electron*, Phys. Rev. **43**, 491 (1933).

The positron is experimentally confirmed!



Take-home messages

1. For heavy atoms ($Z \geq 40$), relativistic effects are essential:
   - ⋆ Scalar effects $\rightarrow$ s, p contraction; d, f expansion
   - ⋆ Spin–orbit coupling $\rightarrow$ $\mathbf{J} = \mathbf{L} + \mathbf{S}$
2. Relativistic QM:
   - ⋆ The Schrödinger is not Lorentz invariant.
   - ⋆ Klein–Gordon is Lorentz invariant but gives negative probabilities (unphysical)
3. Dirac equation:
   - ⋆ Lorentz invariant.
   - ⋆ $\rho \geq 0$
   - ⋆ Predicts spin and positrons.

What I did not cover

Many important aspects omitted for simplicity

1. **Two-electron relativistic interaction (Breit term)**

$$\hat{g}_{\text{Breit}} = -\frac{1}{2r_{12}}\left[\boldsymbol{\alpha}_1 \cdot \boldsymbol{\alpha}_2 + \frac{(\boldsymbol{\alpha}_1 \cdot \mathbf{r}_{12})(\boldsymbol{\alpha}_2 \cdot \mathbf{r}_{12})}{r_{12}^2}\right] \tag{15}$$

2. **Variational collapse of the Dirac Hamiltonian**
3. **Two-component reduction**

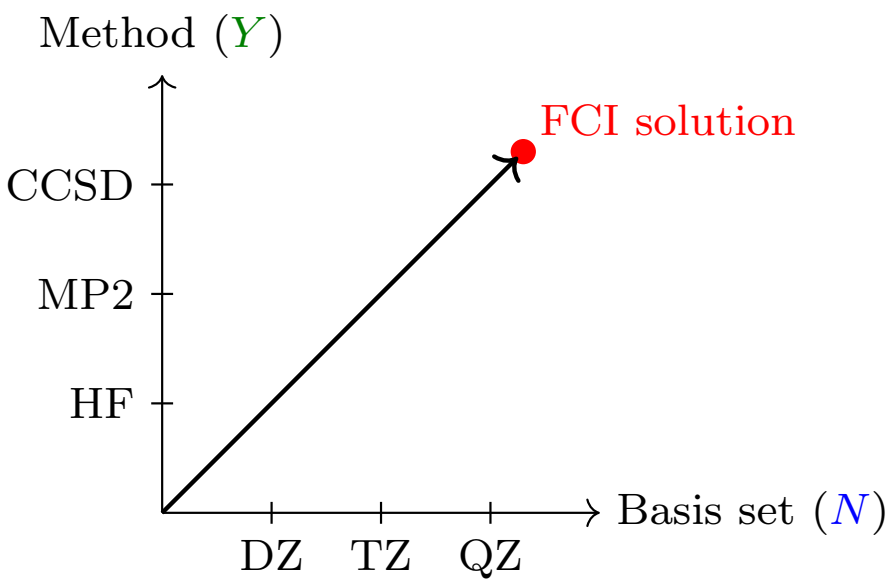
$$\hat{H}_{2c} = \hat{U}^\dagger \hat{H}_{4c} \hat{U} = \begin{pmatrix} H_{++} & 0 \\ 0 & H_{--} \end{pmatrix} \tag{16}$$

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2. Markus Reiher, Alexander Wolf. *Relativistic Quantum Chemistry: The Fundamental Theory of Molecular Science*. Wiley-VCH, 2015.
3. Kenneth G. Dyall, Knut Faegri Jr. *Introduction to Relativistic Quantum Chemistry*. Oxford University Press, 2007.

Theoretical models in chemistry

Hehre et al. (Wiley, 1986); Pople, Rev. Mod. Phys. 71, 1267 (1999).



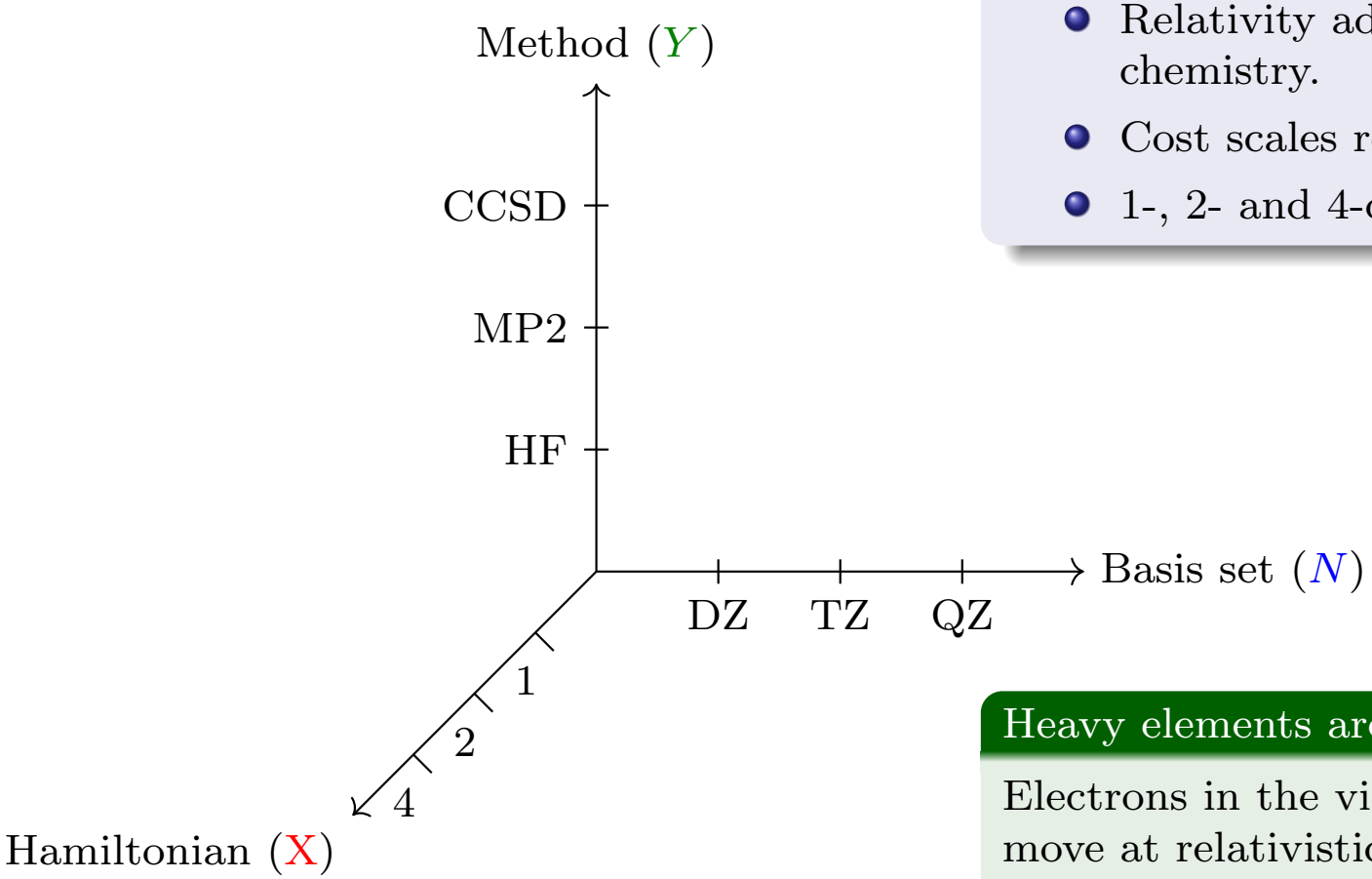
2D Pople Diagram

- Basis set and method are independent.
- Cost scales roughly as $\sim N^Y$.
- FCI: exact solution in a finite basis.

The periodic table is large...

- Relativistic effects are crucial for heavy elements.
- How I'm suppose to include them?

Theoretical models in chemistry



3D Pople diagram

- Relativity adds a new dimension to chemistry.
- Cost scales roughly as $\sim XN^Y$.
- 1-, 2- and 4-component Hamiltonians.

Heavy elements are challenging

Electrons in the vicinity of heavy nuclei move at relativistic speeds.

1. Scalar relativistic effects
2. Spin-orbit coupling

Covariance and contravariance

Why physical laws must be covariant

A contravariant vector transforms as:

$$V'^{\mu} = \frac{\partial x'^{\mu}}{\partial x^{\nu}} V^{\nu} \tag{17}$$

A covariant vector (covector) transforms oppositely:

$$V'_{\mu} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} V_{\nu} \tag{18}$$

Indices are related through the metric:

$$V_{\mu} = g_{\mu\nu} V^{\nu} \tag{19}$$

Covariance of an equation

Its form is preserved under a change of coordinates

$$\text{equation form} \quad \rightarrow \quad \text{same form in any frame.} \tag{20}$$

The laws of physics do not depend on the observer's coordinate system.

Lorentz Transformations

Definition

Lorentz transformations relate space and time coordinates between two inertial frames moving at constant relative velocity v.

1D boost along x

$$x' = \gamma(x - vt), \qquad t' = \gamma\left(t - \frac{v}{c^2}x\right)$$

$$y' = y, \qquad z' = z$$

Key properties

- Speed of light c is invariant
- Space and time mix under boosts
- Reduces to Galilean transformation for $v \ll c$

Physical meaning

Different observers do not agree on distances and times, but agree on spacetime structure.

Why $A_\mu B^\mu$ is Lorentz invariant

- A four-vector transforms under a Lorentz transformation as:

$$A'^\mu = \Lambda^\mu{}_\nu A^\nu, \quad B'^\mu = \Lambda^\mu{}_\nu B^\nu$$

- The scalar product is defined as:

$$A_\mu B^\mu = g_{\mu\nu} A^\mu B^\nu$$

- In a transformed frame:

$$A'_\mu B'^\mu = g_{\mu\nu} (\Lambda^\mu{}_\alpha A^\alpha)(\Lambda^\nu{}_\beta B^\beta)$$

- Using Lorentz invariance of the metric:

$$g_{\mu\nu} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta = g_{\alpha\beta}$$

- Therefore:

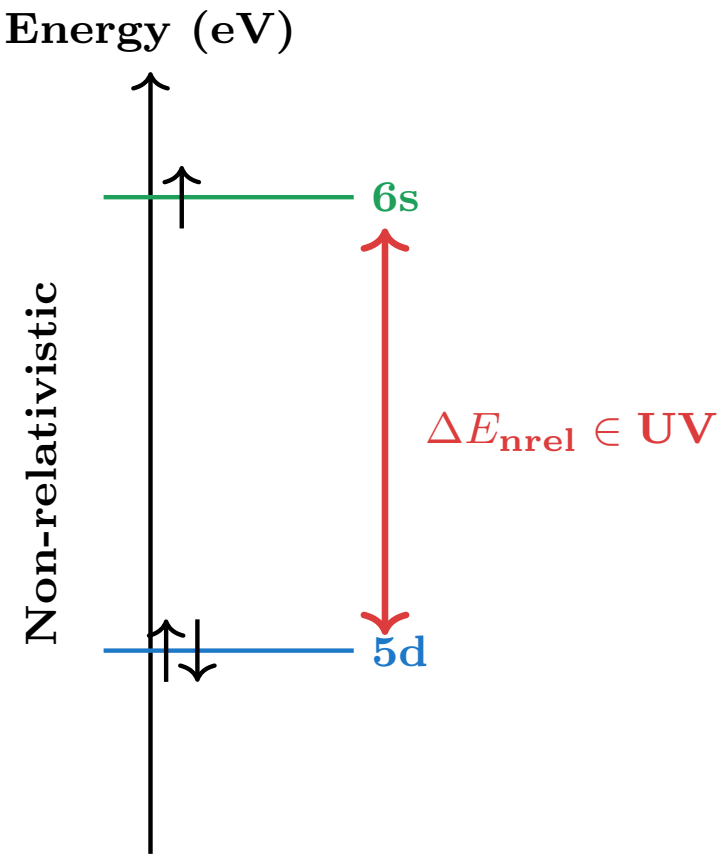
$$A'_\mu B'^\mu = g_{\alpha\beta} A^\alpha B^\beta = A_\mu B^\mu$$

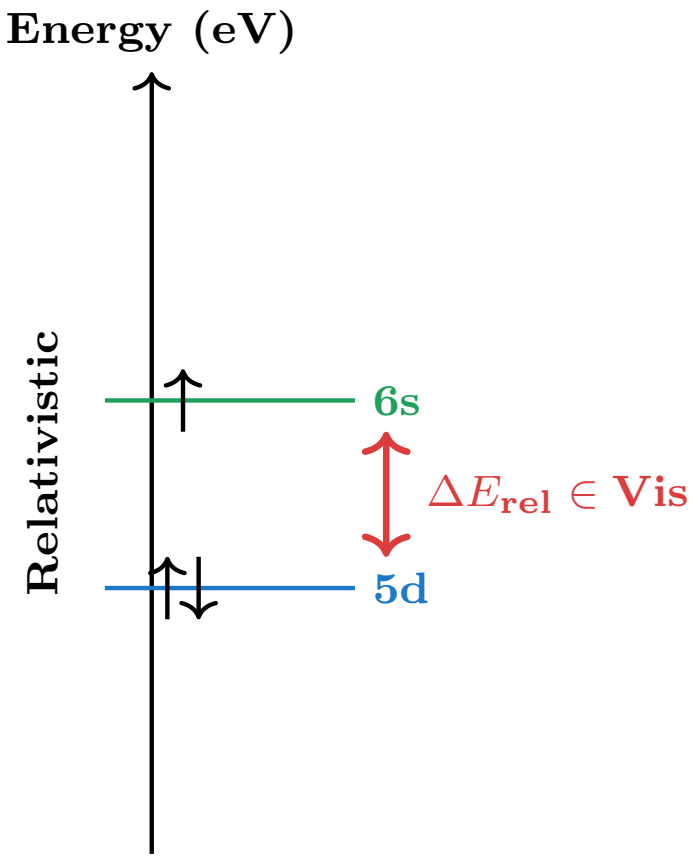
Conclusion: $A_\mu B^\mu$ is Lorentz invariant (a scalar)

Relativistic effects

Scalar effects: why gold is yellow

- Au : $[Xe]\,4f^{14}\,5d^{10}\,6s^{1}$





Relativistic effects

Spin-orbit effect (3)

The Hamiltonian becomes

$$H = H_0 + H_{\mathrm{SO}} \tag{21}$$

Spin-orbit coupling breaks separate conservation of:

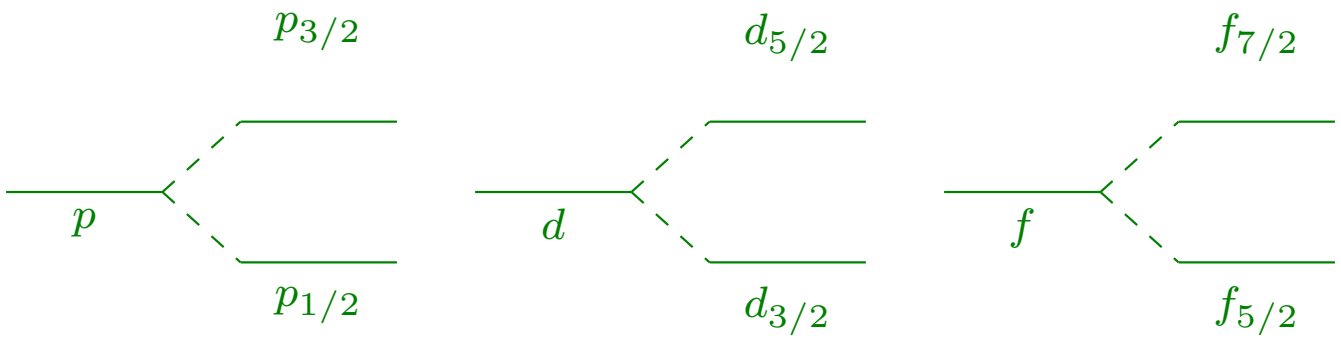
$$[H, L^2] \neq 0, \qquad [H, S^2] \neq 0 \tag{22}$$

However, total angular momentum is conserved:

$$\mathbf{J} = \mathbf{L} + \mathbf{S}, \qquad [H, J^2] = 0 \tag{23}$$

→ J becomes the good quantum number.

→ Origin of the **fine structure**.



Relativistic one-electron part

The Dirac equation (1)

0. A useful identity

$$(\boldsymbol{\sigma}\cdot\mathbf{A})(\boldsymbol{\sigma}\cdot\mathbf{B}) = \mathbf{A}\cdot\mathbf{B} + i\boldsymbol{\sigma}\cdot(\mathbf{A}\times\mathbf{B}) \tag{24}$$

1. Linearization of the KG equation

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - p^2\right)\phi_1 = \left(i\frac{1}{c}\frac{\partial}{\partial t} + \boldsymbol{\sigma}\cdot\mathbf{p}\right)\left(i\frac{1}{c}\frac{\partial}{\partial t} - \sigma\boldsymbol{\sigma}\cdot\mathbf{p}\right)\phi_1 \tag{25}$$

2. The factorization is the product of two coupled equations

$$\left(\frac{i}{c}\frac{\partial}{\partial t} - \boldsymbol{\sigma}\cdot\mathbf{p}\right)\phi_1 = mc\phi_2 \quad (a) \tag{26}$$

$$\left(\frac{i}{c}\frac{\partial}{\partial t} + \boldsymbol{\sigma}\cdot\mathbf{p}\right)\phi_2 = mc\phi_1 \quad (b) \tag{27}$$

Relativistic one-electron part

The Dirac equation (2)

4. Linear combinations $(a+b)$ and $(a-b)$

$$\frac{i}{c}\frac{\partial}{\partial t}(\phi_1+\phi_2)-\boldsymbol{\sigma}\cdot\mathbf{p}(\phi_1-\phi_2)=mc(\phi_1+\phi_2)\quad(a+b)\tag{28}$$

$$-\frac{i}{c}\frac{\partial}{\partial t}(\phi_1-\phi_2)+\boldsymbol{\sigma}\cdot\mathbf{p}(\phi_1+\phi_2)=mc(\phi_1-\phi_2)\quad(a-b)\tag{29}$$

5. Introducing $\psi_{\mathrm{L}}=\phi_1+\phi_2$ and $\psi_{\mathrm{S}}=\phi_1-\phi_2$ we end up with

$$\begin{pmatrix}\frac{i}{c}\frac{\partial}{\partial t} & -(\boldsymbol{\sigma}\cdot\mathbf{p})\\ (\boldsymbol{\sigma}\cdot\mathbf{p}) & -\frac{i}{c}\frac{\partial}{\partial t}\end{pmatrix}\begin{bmatrix}\psi_{\mathrm{L}}\\ \psi_{\mathrm{S}}\end{bmatrix}=\varepsilon\begin{bmatrix}\psi_{\mathrm{L}}\\ \psi_{\mathrm{S}}\end{bmatrix}\tag{30}$$

6. The usual form of the Dirac equation is

$$[\beta mc^2+c(\boldsymbol{\alpha}\cdot\mathbf{p})]\psi=i\frac{\partial}{\partial t}\psi\rightarrow[\beta mc^2+c(\boldsymbol{\alpha}\cdot\mathbf{p})]\psi=\varepsilon\psi\tag{31}$$

$$\alpha_i=\begin{pmatrix}0_2 & \sigma_i\\ \sigma_i & 0_2\end{pmatrix},\qquad\beta=\begin{pmatrix}1_2 & 0_2\\ 0_2 & -1_2\end{pmatrix}\tag{32}$$

The origin of the Darwin term (1)

Zitterbewegung and modified interaction

The rapid relativistic fluctuations of the electron position (*zitterbewegung*) modify the electron–nucleus interaction:

$$V(\mathbf{r}) = -e\varphi(\mathbf{r}) \longrightarrow -e\varphi(\mathbf{r}+\boldsymbol{\delta}) \tag{33}$$

Expanding the potential to second order:

$$\begin{aligned} -e\varphi(\mathbf{r}+\boldsymbol{\delta}) \approx & -e\varphi(\mathbf{r}) - e(\boldsymbol{\delta}\cdot\nabla)\varphi(\mathbf{r}) \\ & - \frac{e}{2}(\boldsymbol{\delta}\cdot\nabla)^2\varphi(\mathbf{r}) \end{aligned} \tag{34}$$

Key idea: the electron samples the potential over a small region of space rather than at a fixed point.

The origin of the Darwin term (2)

Time averaging and final expression

Taking the time average over the rapid fluctuations:

$$V + H_{\mathrm{D}} = -e\varphi(\mathbf{r}) - e\left\langle(\boldsymbol{\delta}\cdot\nabla)\right\rangle_T \varphi(\mathbf{r}) - \frac{e}{2}\left\langle(\boldsymbol{\delta}\cdot\nabla)^2\right\rangle_T \varphi(\mathbf{r}) \tag{35}$$

Assuming isotropy:

$$\langle\delta_i\rangle_T = 0, \qquad \langle\delta_i\delta_j\rangle_T = \frac{1}{3}\langle\delta^2\rangle_T\delta_{ij} \tag{36}$$

$$H_{\mathrm{D}} = -\frac{e}{6}\langle\delta^2\rangle_T\nabla^2\varphi(\mathbf{r}) \tag{37}$$

With fluctuation amplitude:

$$\delta \sim \frac{1}{2mc} \tag{38}$$

Result: the Darwin term corresponds to a *contact correction* arising from spatial averaging of the potential.

The Darwin term for a Coulomb potential

Contact interaction and physical picture

For a Coulomb potential:

$$\varphi(\mathbf{r}) = \frac{Ze}{4\pi\varepsilon_0 r} \tag{39}$$

one obtains:

$$\nabla^2\varphi(\mathbf{r}) \propto \delta(\mathbf{r}), \quad \Rightarrow \quad H_{\mathrm{D}} \propto \delta(\mathbf{r}) \tag{40}$$

The Darwin term is a *contact interaction*, non-zero only at the nucleus.

Physical interpretation (heuristic):

- The strong field near the electron allows virtual e^-e^+ pair creation
- Energy requirement: $\Delta E \sim 2mc^2$
- From uncertainty principle:

$$\Delta E \Delta t \sim 1 \;\Rightarrow\; \Delta t \sim \frac{1}{2mc^2} \tag{41}$$

- Maximum displacement:

$$\Delta x \sim c\Delta t \sim \frac{1}{2mc} \tag{42}$$

Interpretation: the electron position is effectively blurred over a region of size $\sim 1/(2mc)$ due to relativistic quantum fluctuations.